

J. C. Bose University of Science & Technology, YMCA
Department of Computer Engineering
Subject: Basics of Machine Learning (PEC-CS-D-501) **CE52**
Sessional I

Max. Time: 90 mins

Max. Marks: 30

Attempt all questions.

Q1. What is machine learning. How it is different from conventional programming approach? Discuss its learning paradigms. [6],CO1

Q2. Explain bias-variance trade off with diagram. [4],CO1

Q3. Consider the following dataset. Using ID3 find the Entropy and Information Gain value of "Practical Knowledge" attribute. [10],CO2

CGPA	Interactive	Practical Knowledge	Skills	Job Offer
≥ 9	Yes	Very good	Good	Yes
> 8	No	Good	Moderate	Yes
< 8	No	Average	Poor	No
< 8	No	Average	Good	No
> 8	Yes	Good	Moderate	Yes
> 8	Yes	Good	Poor	No
≥ 9	Yes	Good	Moderate	Yes
≥ 9	Yes	Very good	Good	Yes
> 8	Yes	Good	Good	Yes
< 8	Yes	Average	Good	No

Q4. Fit a linear regression line for data. Predict Y for X=8. Also calculate the error using Least Square method and plot the graphs. To

X	1	2	3	4	5
Y	2	4	5	4	5

[10],CO2